

HAMZA BASHARAT

Artificial Intelligence / Machine Learning Engineer

Lahore, Pakistan | +92 300 6547302 | hamzabasharat2004@gmail.com

linkedin.com/in/hamzabasharat26 | github.com/hamzabasharat26 | linkedin.com/in/hamzabasharat26/details/recommendations

PROFESSIONAL SUMMARY

BS Computer Engineering graduate with hands-on experience designing, developing, and deploying AI/ML models in Python using PyTorch, TensorFlow, and Scikit-learn. Practical exposure to Machine Learning algorithms, Computer Vision, NLP, and Generative AI through internship, contract, and academic projects spanning data preprocessing, feature engineering, model evaluation, model optimization, and performance tuning. Comfortable with SQL, Git, and cloud platforms (AWS, Azure), with strong analytical and problem-solving skills. Eager to collaborate with senior engineers to build scalable AI solutions in a fast-paced, innovation-driven environment.

TECHNICAL SKILLS

Programming: Python (primary), C++, C, JavaScript, SQL, MATLAB; data structures and algorithms, object-oriented programming, probability and statistics

Machine Learning: Supervised and unsupervised learning, regression, classification, clustering, CNNs, transformers, data preprocessing, feature engineering, exploratory data analysis, model training and evaluation, hyperparameter tuning, model optimization, performance tuning

Libraries & Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, Hugging Face Transformers and Diffusers

Computer Vision, NLP & Generative AI: Object detection (YOLOv8/v11), classification, segmentation, pose estimation, anomaly detection, Vision Transformers (ViT), OCR, text embeddings, RAG pipelines, LangChain, LangGraph, vector databases (FAISS, Chroma), LLM prompt engineering and fine-tuning, diffusion models, OpenAI and Anthropic Claude APIs

Cloud, Data & Tools: AWS (EC2, S3), Microsoft Azure AI, Docker, ONNX, FastAPI, Flask, Streamlit, MySQL, MongoDB, Git and GitHub (version control), Linux, Jupyter Notebook, technical documentation

PROFESSIONAL EXPERIENCE

AI & Machine Learning Engineer | Robionix Technologies | Islamabad, Pakistan | Aug 2024 - Jul 2026

- Built and deployed an automated garment measurement platform processing 500+ items per day at 90%+ accuracy, by training PyTorch computer vision models and serving them as a documented REST API with FastAPI and Docker on AWS EC2.
- Raised detection and pose-estimation accuracy to 92%+ mAP at 15+ FPS on constrained hardware by running structured experimentation across data augmentation, feature engineering, hyperparameter tuning, and model optimization.
- Reduced manual data entry in quality-control reporting by building an OCR extraction pipeline that parses labels and documents into structured MySQL records for downstream analysis and insights.
- Developed Generative AI and NLP assistants with LangChain, LangGraph, and LLM APIs over FAISS and Chroma vector databases, and closed dataset class-imbalance gaps by generating synthetic data with diffusion models and fine-tuning Vision Transformers on Hugging Face.

AI Engineer (Contract) | Essenceware Technologies | Pakistan | Jul 2025 - Sep 2025

- Delivered a real-time PPE and workplace-safety detection model end to end, from dataset cleaning and preprocessing through evaluation to a Flask and Streamlit inference dashboard with model versioning, documenting training runs and results for internal handoff.

AI Engineer, Intern | NESCOM, National Development Complex | Islamabad, Pakistan | Jun 2025 - Sep 2025

- Developed UAV disaster-response perception pipelines in Python and OpenCV, validated through Gazebo simulation and integration testing, and presented results to government stakeholders evaluating the system for national emergency preparedness.

ACADEMIC & AI/ML PROJECTS

- **AI-Enabled Fabric Defect Detection (Final Year Project):** Trained a YOLO classification model to mAP50 of 0.81 on a self-curated 1,400+ image dataset and fused it with PatchCore anomaly detection on an edge camera, adding severity grading and automated CSV/JSON reporting. Code: github.com/hamzabasharat26/Automated-Fabric-Defect-Detection-System
- **CV Recruiter RAG Agent (NLP and Generative AI):** Resume-screening agent that parses PDF and DOCX files into a FAISS vector store and ranks candidates against a job description with citation-backed scoring; LangGraph, LLM APIs, FastAPI.
- **Dock Vision AI, 1st Place IEEE Hackathon 2024:** Real-time vessel-docking safety system using video analytics for spatial localization and alignment detection; converted into a paid commercial deployment. Code: github.com/hamzabasharat26/DockVision-AI

EDUCATION

Bachelor of Science in Computer Engineering | National University of Technology (NUTECH), Islamabad, Pakistan | 2022 - 2026

Relevant coursework: Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Deep Learning, Digital Image Processing, Probability and Statistics, Database Systems.

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Machine Learning Engineer - Associate, Amazon Web Services | Microsoft AI & ML Engineering Professional Certificate and Azure AI / Computer Vision, Microsoft
- Advanced Computer Vision with TensorFlow and LangChain for LLM Application Development, DeepLearning.AI | Exploratory Data Analysis for Machine Learning, IBM
- Programming for Everybody (Getting Started with Python), University of Michigan, Feb 2025 | Introduction to Embedded Machine Learning, Edge Impulse, Mar 2025
- Robotics, National Vocational and Technical Training Commission (NAVTTTC), Aug 2023 | Claude 101 and AI Fluency Framework Foundations, Anthropic
- **Awards:** 1st Place, IEEE Hackathon 2024 | 1st Place, ICAT National Robotics Competition | Runner-Up, National AI Sprint (NAIS) 2025, National Centre of Physics